INCH-POUND
MIL-PRF-19978/21B
9 December 2021
SUPERSEDING
MIL-C-19978/21A
18 June 2001

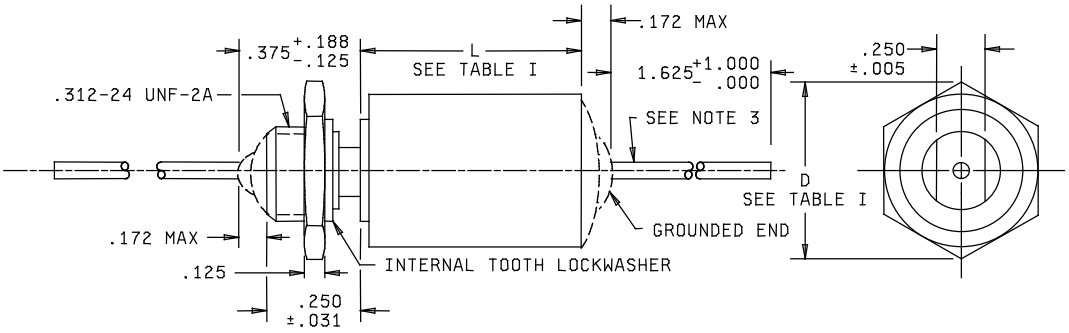
PERFORMANCE SPECIFICATION SHEET

# CAPACITOR, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, THREADED-STUD RETAINER, TUBULAR (UNINSULATED), (HERMETICALLY SEALED IN METAL CASES), STYLE CQ11

INACTIVE FOR NEW DESIGN
AFTER 30 AUGUST 1974

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19978.




| Inches | mm |
|---|---|
| .001 | 0.03 |
| .004 | 0.10 |
| .005 | 0.13 |
| .031 | 0.79 |
| .032 | 0.81 |
| .125 | 3.18 |
| .172 | 4.37 |
| .250 | 6.35 |
| .312 | 7.92 |
| .375 | 9.53 |
| 1.000 | 25.40 |
| 1.625 | 41.28 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. Leads shall be of tinned solid wire, .032 inch + .004 inch, -.001 inch (No. 20 AWG).
4. Mounting nuts and lockwashers shall be supplied.

FIGURE 1. Style CQ11 capacitors.

AMSC N/A
FSC 5910
DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

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REQUIREMENTS:

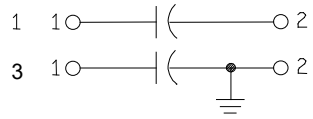
Dimensions and configuration: See figure 1 and table I.

Case material: Nonmagnetic (screwneck and end seal may be of magnetic material).

Dielectric material: Paper and polyethelene terephthalate (K).

Terminals: Axial-wire lead (see figure 1).

Circuit diagram:




Rated voltage: See table I.

Rated temperature: -65° to +125°C.

Capacitance (nominal): See table I.

Capacitance tolerance: ± 10 percent (symbol K).

Vibration grade: 3.

Dissipation factor (DF) (max): 1.0 percent.

Seal: MIL-STD-202-112, test condition letter A.

Barometric pressure, qualification only: In accordance with MIL-PRF-19978.

Test points:

Circuit diagram 1: Between terminals and case.

Circuit diagram 3: Between ungrounded terminal and case.

Vibration, high frequency: MIL-STD-202-204, test condition B, with the following exception:

Direction and duration of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt atmosphere: In accordance with MIL-PRF-19978.

Solderability: In accordance with MIL-PRF-19978.

Terminal strength: MIL-STD-202-211, test condition letter D.



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TABLE I. Style CQ11 capacitors.

| Part or identifying number (PIN) 1/ | DC voltage rating (volts) | Capacitance (µF) | Case dimensions (inches) 2/ | | |
|---|---|---|---|---|---|
| | | | | Circuit 1 | Circuit 3 |
| | | | D+.010, -.020 | L ± .015 | L ± .015 |
| CQ11A-KB333K3 | 100 | .033 | 0.400 | 0.771 | 0.708 |
| CQ11A-KB154K3 | 100 | .15 | 0.400 | 0.896 | 0.833 |
| CQ11A-KB224K3 | 100 | .22 | 0.400 | 1.146 | 1.083 |
| CQ11A-KB394K3 | 100 | .39 | 0.562 | 1.156 | 1.094 |
| CQ11A-KB474K3 | 100 | .47 | 0.562 | 1.156 | 1.094 |
| CQ11A-KB684K3 | 100 | .68 | 0.562 | 1.406 | 1.344 |
| CQ11A-KB105K3 | 100 | 1.0 | 0.562 | 1.906 | 1.844 |
| CQ11A-KC683K3 | 200 | .068 | 0.400 | 0.771 | 0.708 |
| CQ11A-KC104K3 | 200 | .10 | 0.400 | 0.896 | 0.833 |
| CQ11A-KC154K3 | 200 | .15 | 0.400 | 1.146 | 1.083 |
| CQ11A-KC224K3 | 200 | .22 | 0.400 | 1.396 | 1.333 |
| CQ11A-KC334K3 | 200 | .33 | 0.562 | 1.156 | 1.094 |
| CQ11A-KC474K3 | 200 | .47 | 0.562 | 1.406 | 1.344 |
| CQ11A-KC684K3 | 200 | .68 | 0.562 | 1.906 | 1.844 |
| CQ11A-KC105K3 | 200 | 1.0 | 0.670 | 1.906 | 1.844 |
| CQ11A-KE103K3 | 400 | .010 | 0.400 | 0.771 | 0.708 |
| CQ11A-KE393K3 | 400 | .039 | 0.400 | 0.896 | 0.833 |
| CQ11A-KE473K3 | 400 | .047 | 0.400 | 0.896 | 0.833 |
| CQ11A-KE683K3 | 400 | .068 | 0.400 | 1.146 | 1.083 |
| CQ11A-KE104K3 | 400 | .10 | 0.400 | 1.396 | 1.333 |
| CQ11A-KE154K3 | 400 | .15 | 0.562 | 1.156 | 1.094 |
| CQ11A-KE224K3 | 400 | .22 | 0.562 | 1.406 | 1.344 |
| CQ11A-KE274K3 | 400 | .27 | 0.562 | 1.906 | 1.844 |
| CQ11A-KE334K3 | 400 | .33 | 0.562 | 1.906 | 1.844 |
| CQ11A-KE474K3 | 400 | .47 | 0.670 | 1.906 | 1.844 |
| CQ11A-KE684K3 | 400 | .68 | 0.750 | 2.156 | 2.094 |
| CQ11A-KE105K3 | 400 | 1.0 | 1.000 | 1.906 | 1.844 |
| CQ11A-KF102K3 | 600 | .001 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF152K3 | 600 | .0015 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF222K3 | 600 | .0022 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF332K3 | 600 | .0033 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF472K3 | 600 | .0047 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF682K3 | 600 | .0068 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF103K3 | 600 | .010 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF153K3 | 600 | .015 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF223K3 | 600 | .022 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF333K3 | 600 | .033 | 0.400 | 0.896 | 0.833 |
| CQ11A-KF473K3 | 600 | .047 | 0.400 | 1.146 | 1.083 |
| CQ11A-KF683K3 | 600 | .068 | 0.400 | 1.396 | 1.333 |
| CQ11A-KF104K3 | 600 | .10 | 0.562 | 1.156 | 1.094 |
| CQ11A-KF154K3 | 600 | .15 | 0.562 | 1.406 | 1.344 |
| CQ11A-KF184K3 | 600 | .18 | 0.562 | 1.906 | 1.844 |
| CQ11A-KF224K3 | 600 | .22 | 0.562 | 1.906 | 1.844 |
| CQ11A-KF334K3 | 600 | .33 | 0.670 | 1.906 | 1.844 |
| CQ11A-KF474K3 | 600 | .47 | 0.750 | 2.156 | 2.094 |
| CQ11A-KF684K3 | 600 | .68 | 1.000 | 1.906 | 1.844 |

1/ Complete part number shall include an additional symbol to indicate circuit.
2/ Dimensions are in inches. See table II for metric equivalents.



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TABLE II. Millimeter equivalent of decimal inches.

| Inches | mm | Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|---|---|
| 0.010 | 0.25 | 0.670 | 17.02 | 0.896 | 22.76 | 1.156 | 29.36 | 1.844 | 46.84 |
| 0.015 | 0.38 | 0.708 | 17.98 | 1.000 | 25.40 | 1.333 | 33.86 | 1.906 | 48.41 |
| 0.020 | 0.51 | 0.750 | 19.05 | 1.083 | 27.51 | 1.344 | 34.14 | 2.094 | 53.19 |
| 0.400 | 10.16 | 0.771 | 19.58 | 1.094 | 27.79 | 1.396 | 35.46 | 2.156 | 54.76 |
| 0.562 | 14.27 | 0.833 | 21.16 | 1.146 | 29.11 | 1.406 | 35.71 | | |

Insulation resistance (IR):

Terminal to terminal: See table III.

Terminal to case: Greater than 10,000 megohms.

TABLE III. Terminal-to-terminal insulation resistance.

| Capacitance rating | Minimum insulation resistance |
|---|---|
| | At +25°C |
| 0 to 0.6 microfarad | 25,000 megohms |
| Greater than 0.6 microfarad | 15,000 megohm-microfarads 1/ |
| | At +125°C |
| 0 to 0.08 microfarad | 250 megohms |
| Greater than 0.08 microfarad | 20 megohm-microfarads 1/ |

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Stability at low and high temperature:

Low temperature:

Test temperature: -65°C +0°C, -5°C.

Capacitance change (max): -10 percent.

High temperature:

Test temperature: +125°C +5°C, -0°C.

Capacitance change (max): +10 percent.

Life:

Capacitance change (max): ± 5 percent.

Resistance to soldering heat: In accordance with MIL-PRF-19978.



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NOTE: Style CQ11 capacitors are substitutable for style CP11 capacitors formerly covered in MIL-C-25/1K as shown in the following:

| From MIL-C-25/1K (CP11) | To MIL-PRF-19978/21 (CQ11) | From MIL-C-25/1K (CP11) | To MIL-PRF-19978/21 (CQ11) |
|---|---|---|---|
| CP11A-KB333K- | CQ11A-KB333K3- | CP11A-KE334K- | CQ11A-KE334K3- |
| CP11A-KB154K- | CQ11A-KB154K3- | CP11A-KE474K- | CQ11A-KE474K3- |
| CP11A-KB224K- | CQ11A-KB224K3- | CP11A-KE684K- | CQ11A-KE684K3- |
| CP11A-KB394K- | CQ11A-KB394K3- | CP11A-KE105K- | CQ11A-KE105K3- |
| CP11A-KB474K- | CQ11A-KB474K3- | CP11A-KF102K- | CQ11A-KF102K3- |
| CP11A-KB684K- | CQ11A-KB684K3- | CP11A-KF152K- | CQ11A-KF152K3- |
| CP11A-KB105K- | CQ11A-KB105K3- | CP11A-KF222K- | CQ11A-KF222K3- |
| CP11A-KC683K- | CQ11A-KC683K3- | CP11A-KF332K- | CQ11A-KF332K3- |
| CP11A-KC104K- | CQ11A-KC104K3- | CP11A-KF472K- | CQ11A-KF472K3- |
| CP11A-KC154K- | CQ11A-KC154K3- | CP11A-KF682K- | CQ11A-KF682K3- |
| CP11A-KC224K- | CQ11A-KC224K3- | CP11A-KF103K- | CQ11A-KF103K3- |
| CP11A-KC334K- | CQ11A-KC334K3- | CP11A-KF153K- | CQ11A-KF153K3- |
| CP11A-KC474K- | CQ11A-KC474K3- | CP11A-KF223K- | CQ11A-KF223K3- |
| CP11A-KC684K- | CQ11A-KC684K3- | CP11A-KF333K- | CQ11A-KF333K3- |
| CP11A-KC105K- | CQ11A-KC105K3- | CP11A-KF473K- | CQ11A-KF473K3- |
| CP11A-KE103K- | CQ11A-KE103K3- | CP11A-KF683K- | CQ11A-KF683K3- |
| CP11A-KE393K- | CQ11A-KE393K3- | CP11A-KF104K- | CQ11A-KF104K3- |
| CP11A-KE473K- | CQ11A-KE473K3- | CP11A-KF154K- | CQ11A-KF154K3- |
| CP11A-KE683K- | CQ11A-KE683K3- | CP11A-KF184K- | CQ11A-KF184K3- |
| CP11A-KE104K- | CQ11A-KE104K3- | CP11A-KF224K- | CQ11A-KF224K3- |
| CP11A-KE154K- | CQ11A-KE154K3- | CP11A-KF334K- | CQ11A-KF334K3- |
| CP11A-KE224K- | CQ11A-KE224K3- | CP11A-KF474K- | CQ11A-KF474K3- |
| CP11A-KE274K- | CQ11A-KE274K3- | CP11A-KF684K- | CQ11A-KF684K3- |

Referenced documents: In addition to MIL-PRF-19978, this document references the following:

MIL-STD-202-112
MIL-STD-202-204
MIL-STD-202-211

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Review activities:
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